

Vehicle Telematics & Trip Intelligence Platform

Product Vision, Hardware Ingestion Pipeline, Financial Models, and Full Interface Specification

Author:

Platform Engineering & UI Systems

Hardware Target:

WiCAN Pro (ESP32-S3 OBD-II CAN-Bus)

Self-Hosted Server:

Docker Hub on LAN / Tailscale

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1. Executive Summary: What is the Point of this App?

1.1 The Fundamental Problem

Modern automobiles generate millions of high-fidelity data frames across their Controller Area Network (CAN-bus) every minute: battery state-of-charge, fuel injector volume, engine coolant temperature, wheel RPM, and regenerative braking energy recapture.

Despite this wealth of raw telemetry, consumer vehicle ownership software remains fundamentally broken:

1. **Subscription Gating & OEM Lock-in:** Manufacturers gate basic remote telemetry behind monthly fees (\$10–\$25/month) while mining and selling personal driving habits to insurance data brokers.
2. **Inaccurate Financial Tracking:** Existing fuel apps (like Fuelly or GasBuddy) rely on manual odometer logging at gas stations and completely fail on modern dual-fuel Plug-in Hybrids (PHEVs) and Electric Vehicles (EVs). They cannot track how much money was spent on electricity versus gasoline for any specific trip.
3. **Distracting In-Car Interfaces:** Most DIY OBD-II apps (Torque Pro, OBD Fusion) are designed as live diagnostic HUDs meant to be used on windshield mounts while driving, creating cognitive distraction and draining phone batteries.
4. **Cryptic Mechanic Tools:** Traditional OBD scanners throw raw hexadecimal Diagnostic Trouble Codes (DTCs) and freeze-frame sensor graphs that terrify non-mechanic owners without explaining whether the car is safe to drive.

1.2 The Platform Mission

This application is a **privacy-first, autonomous automotive telematics, trip intelligence, and energy expense tracking platform**. It bridges the gap between hardware engineering and elegant consumer mobile software.

Core Architectural Pillars

- **Autonomous Post-Drive Ingestion:** The app is strictly designed for post-drive review. The in-vehicle hardware (WiCAN Pro) wakes up, logs CAN packets to internal flash while driving, and silently syncs over home Wi-Fi when parked. Zero live phone connection required.
- **True Multi-Fuel Financial Accounting:** Calculates the exact dollar cost of every trip, split between electricity and gasoline, using a single **Energy Lot Ledger** for every energy source (\$3.1).
- **Owner-First Diagnostics:** Strips away mechanic PID clutter. Tells the owner in plain English what system is complaining, when it was detected, during which trip, and whether the vehicle is safe to drive.
- **Zero Main Nav / Button-Driven Architecture:** Eliminates persistent bottom tab bars and hamburger drawers. Every view is accessed via direct, contextual card touchpoints with intuitive back-button navigation.

1.3 v1 Launch Scope

- **Market:** United States only. Road-load coefficients (EPA Test Car List) and fallback fuel prices (EIA) are US data sources. Other regions are a roadmap item (\$6).
- **Clients:** Android app (Kotlin + Jetpack Compose) and a web interface served by the home server. No native iOS app in v1; iOS users can use the web interface.
- **Hardware:** WiCAN Pro. Other scanners via the phone relay are a roadmap item (\$2.5).

2. Hardware & Ingestion Architecture

2.1 The Autonomous WiCAN Pro Lifecycle

The platform relies on the **WiCAN Pro** (an open-source ESP32-S3 automotive scanner plugged directly into the standard OBD-II port below the steering column). The hardware operates completely independently of the driver's phone:

Vehicle State	WiCAN Pro Behavior	Data & Power Impact
1. Ignition ON	Wakes on CAN-bus voltage activity (> 13.1V or bus activity). Initializes SD/SPI Flash circular buffer.	Draws 120mA. Zero interaction needed from driver.
2. Driving	Autonomously logs the vehicle profile's signals: Speed, Odometer, SoC%, Fuel Rate, DTCs, and GPS position when the vehicle broadcasts it (see §2.4).	100% offline. Zero cellular data plan or phone pairing required.
3. Ignition OFF	Detects engine stop / CAN bus sleep. Closes active trip file. Enters ultra-low power sleep.	Sub-1mA parasitic battery draw (safe for months of parking).
4. Parked at Home	Wakes Wi-Fi radio on home garage network. Batches and pushes encrypted CAN trip logs to Docker server.	Fast sync (10–30 seconds) via local HTTP/MQTT POST.
5. Server Process	Home server decodes the logged signals using the vehicle profile (§2.3), calculates financial costs, updates vehicle garage, and flags DTCs.	Data stored in local PostgreSQL / SQLite database under owner's control.

2.2 Privacy & Local Network Independence

The self-hosted Docker backend (Home Telematics Hub) runs locally on the owner's home server (e.g. Unraid, TrueNAS, Raspberry Pi, or Synology) and is accessible remotely via secure Tailscale or WireGuard tunnels.

In self-hosted mode, no third-party cloud server ever touches the vehicle's telemetry or location history. A future hosted plan is strictly opt-in and must state exactly what data it stores and why.

OBD-II First, Cloud-Optional Design

The user interface is engineered to look 100% complete and fully featured relying exclusively on OBD-II / CAN-bus telemetry. Optional Home Assistant or OEM cloud integrations (e.g. remote door lock or cabin pre-conditioning) are treated as progressive modular enhancements that gracefully hide without leaving awkward empty gaps if unavailable.

2.3 Vehicle Compatibility Layers

A **vehicle profile** tells the logger which signals to read and how to decode them. Profiles are assembled from three layers, in order of reliability:

Layer	What It Provides	Caveats
1. Generic OBD-II (SAE J1979)	Speed (010D), RPM, mass airflow (0110), fuel level (012F), barometric pressure (0133), ambient temperature, odometer (01A6) where supported. Works on essentially every US car from 2008 onward.	No standard battery current or SoC on most EVs. Fuel rate (015E) is rarely supported, so fuel burned is usually derived from mass airflow (§3.4).
2. Community poll lists (Mode 22)	Manufacturer-specific values such as HV pack voltage and current, SoC, and heater power. Sources: OVMS vehicle modules, WiCAN vehicle profiles, ABRP EV PID lists.	Coverage varies by model. Check each source's license before bundling. Every entry must be verified on a real vehicle before it is marked verified.
3. Broadcast CAN (opendbc DBC files)	Passively decoded broadcast messages, e.g. ford_lincoln_base_pt.dbc includes PHEV battery, charge status, and GPS messages. MIT licensed.	Written for openpilot, which taps behind the camera rather than at the OBD port. Many newer cars gateway broadcast traffic away from the OBD port. Only use messages confirmed visible at the port.

The 2017 Ford Fusion Energi is the reference vehicle. Its verified signals so far are HV pack voltage (224807) and pack current (22480B) from the BECM; FORScan's row indices (e.g. "1456") are **not** request identifiers and must never be used as PIDs.

2.4 Position & Elevation Sources

GPS is an enhancement, never a requirement. Trip cost and the core energy ledger work without it. Sources are tried in this order:

1. **Vehicle broadcast GPS:** Some infotainment units broadcast position on CAN. Ford's base DBC defines APIMGPS_Data_Nav_1/2/3 (position, speed, heading, altitude); the Fusion's SYNC 3 unit likely sends equivalent frames, which must be confirmed by sniffing.
2. **GPS receiver on the scanner:** For vehicles that don't broadcast position, a GPS module connected to the scanner is logged with the same clock as the CAN data, so no merge is needed. Whether the WiCAN Pro accepts an external receiver must be confirmed before this is offered.
3. **No position:** Maps and route features are hidden. The phone's location is **never** used: the app does not request location permission, and trip positions come only from the vehicle or scanner. Elevation falls back to barometric pressure (0133).

Elevation, when a position is known, comes from vehicle GPS altitude, else from an open terrain elevation dataset looked up by position, else from barometric pressure.

2.5 Scanner Support & Phone Relay (Roadmap)

The WiCAN Pro is the recommended scanner because it logs and uploads without a phone. Other scanners (e.g. Bluetooth ELM327-class adapters such as OBDLink or Veepeak) cannot reach the server themselves. On the roadmap, the mobile app relays for them: it connects over Bluetooth during the drive, buffers the log on the phone, and uploads when the phone reaches the home network. The phone relay reuses the same vehicle profiles and trip format as §2.3–2.4. The relay does not record the phone's location.

3. Cost, Energy & UX Models

3.1 The Energy Lot Ledger (Trip Cost Accounting)

Every energy store on the vehicle (a fuel tank, a plug-in traction battery) is tracked by one shared accounting engine. Gasoline and electricity use the same rules, so a pure EV, a PHEV, and a gas car differ only in which stores they have.

Lots

Each fill-up or charge session creates a **lot**: the energy added to the store (gallons, or kWh stored in the battery), its total price, and where it happened. Driving consumes lots **oldest first** (FIFO). A trip's cost is the value of the lot energy it consumed.

$$C_{\text{trip}} = \sum_{\text{lots}} E_{\text{drawn from lot}} \cdot P_{\text{lot}}$$

Where a Lot's Amount Comes From

A lot's size is never read off the gauge when a better figure exists. Fuel-level senders are coarse, move with slope and temperature, and are nonlinear near full; SoC is a model inside the battery controller. So the gauge only **signals** that a refill happened. The amount comes from one of:

Source	Used For
Entered	Fill-ups and paid public charges. The user enters exactly two things from the receipt: the amount (gallons or kWh) and the total cost. This is the only manual step in the app.
Measured	Home and free charging. When the scanner is awake during charging, it integrates pack voltage × current for the whole session and reports the kWh that went into the battery, with start and end times. No user input: the price is the Home Electricity Rate at Home, \$0.00 at a saved free-charging place.
Gauge (approximate)	Only when nothing better exists: a detected fill-up whose receipt has not been entered yet, or a charge the scanner slept through. The amount is the level rise while parked. Every trip that draws from such a lot is labeled approximate.

Detecting Fill-Ups and Charges

- **Between-trip comparison (always works):** when a trip starts, its fuel level and SoC are compared with the end of the previous trip. A rise while parked (fuel ≥ 2 gal, or battery ≥ 2% SoC) means a refill happened, even if the scanner slept through it.
- **Charge sessions:** a measured session from the scanner replaces any gauge-detected charge in the same parked window, so the same charge is never counted twice.
- **Location:** the lot takes the parking position from the end of the previous trip; the phone's location is never used (§2.4).
- **Details prompt:** a detected fill-up, or a charge at a place that is neither Home nor a free place, shows `Fill-up detected • Add details` until the user enters the amount and cost. The gauge's figure is prefilled and labeled as the gauge's, so the user can confirm it against the receipt. Trips drawing from it show that they are waiting on the receipt; entering it re-costs them.

No Resets to the Gauge

Trips consume their **measured** energy (pack power and fuel rate summed over the drive, §3.4), and lots come from receipts and measured sessions. The ledger never adjusts itself to match a gauge reading: doing so would feed the gauge's own errors into costs. Because both sides are measured, the ledger's running total and the gauge agree within the gauge's accuracy without any correction. The gauge has no say in any cost.

Exact Zeros and Exact Totals

- **Free energy is exactly \$0.** Free charging (e.g. a workplace or school charger) is an explicit \$0.00 rate tied to a saved location, never an “unpriced” entry. Once a paid lot is used up, trips drawing only from free lots cost exactly \$0.00 and are labeled with the source (e.g. Free • School charger).
- **Dust sweep:** a lot with less than 0.1 kWh (or 0.02 gal) left is closed, and its remaining value is charged to the current trip. Oldest-first lots already empty completely, so this only absorbs measurement residue.
- **Money is exact:** the ledger works in millionths of a dollar and rounds each trip to the cent with a running total, so trip cents always add up exactly.
- **Reconciliation invariant:** the sum of all trip costs plus the value still in the tank and battery equals the sum of all money paid, to the cent.

Pricing a Lot

- **Electricity:** lot price is the total paid divided by kWh **stored in the battery**, so charger losses are included. For a measured home charge, the wall energy is the charger’s own reading when it reports one; otherwise the stored kWh divided by a configurable charger efficiency (default 88% Level 2, 80% Level 1), at the Home Electricity Rate.
- **Gasoline:** the receipt’s total cost for the receipt’s gallons.
- **Before the receipt is entered:** a fill-up is valued at the U.S. EIA weekly regional retail gasoline price for that date, then the Baseline Gasoline Price in Settings; an unpriced charge at the Home Electricity Rate. These stand-ins are marked and replaced as soon as the receipt is entered.

Powertrain Behavior

Powertrain	Stores in the Ledger
Pure EV	Traction battery only.
PHEV	Fuel tank and traction battery. Each trip draws from both according to measured consumption.
HEV	Fuel tank only. The hybrid battery is an internal buffer charged by the engine and regen. A trip that ends with a different hybrid SoC than it started borrowed or banked energy; that net change is converted to fuel at the trip’s measured efficiency so costs are not shifted between trips.
Gas (ICE)	Fuel tank only.

Why Not an Average of Recent Fill-Ups or a Tank Mixture?

An average of the last few fill-ups treats a small top-up as heavily as a full tank and ignores fuel already burned. A physical tank-mixture model is more realistic but never fully forgets a lot: after one paid public charge followed by free charging, trips would show costs like \$0.0001 forever. FIFO lots give intuitive answers, exact zeros, and totals that reconcile to the cent.

3.2 Dual-Fuel Powertrain Consistency

The platform natively handles all four modern powertrain architectures with tailored visual telemetry:

Powertrain	Accent Color		Card & Gauge Architecture
Pure EV (Volvo EX30)	Emerald (#0C9660)	Green	Full-width Battery SoC card (82% • 210 mi) with solid progress bar. Line 3 financials: 8.2 kWh (4.1 mi/kWh). No ghost fuel metrics.
PHEV (Ford Fusion)	Electric (#1E6FD9)	Blue	Dual sub-cards (Electric % + Gas %) with dual progress bars. Each trip shows its fuel-free share.
HEV (Toyota RAV4)	Warm (#D97706)	Amber	Full-width Fuel Tank card (76% • 420 mi). Line 3 financials: 42.5 mpg • 52% fuel-free. Fuel-free miles and regen on the trip page.
ICE (Volvo XC90)	Slate (#374151)	Charcoal	Full-width Fuel Tank card (76% • 320 mi). Line 3 financials: 24.1 mpg (2.4 gal). Zero electric artifacts.

3.3 The Majority Energy Squirrel System

On PHEV drive lists, transactions are represented by 44 × 34 px **squircles** acting as visual transaction avatars:

- **Majority Electric (> 50% EV):** Bold vector bolt on an Electric Blue-dominant linear gradient (60% Blue fading to 40% Slate).
- **Majority Gas (> 50% Gas):** Bold vector pump on a Slate-dominant linear gradient (25% Blue fading to 75% Slate).
- **Pure Electric (100% EV):** Bold vector bolt on a solid soft Electric Blue tint squiracle (#1E6FD9 0.14).

3.4 Data Availability Tiers & Graceful Degradation

Most vehicles will not expose every value. Each trip is computed from the best source available, and the UI states which one was used. Nothing is displayed that the data cannot support.

Energy In (What the Trip Consumed)

Tier	Source	Typical Availability
A. Measured	Battery power from pack voltage $\times$ current; fuel from a fuel-rate signal. Second-by-second.	Vehicles with a verified manufacturer profile (e.g. Fusion Energi).
B. Derived	Fuel burned from mass airflow ($\dot{m}_{\text{fuel}} = \dot{m}_{\text{air}}/14.7$, adjusted by fuel trims); speed-density engines use MAP, RPM, and intake temperature instead. Battery energy from SoC change $\times$ usable capacity.	Nearly all gas cars (mass airflow or MAP is standard). EVs with an SoC signal from community lists.
C. Refill-Based	Energy bought between two refills is spread across the trips in between, in proportion to each trip's distance, and labeled as such.	Every vehicle, as long as fill-ups or charges are logged.

Motion & Terrain

- **Speed** (010D) is available on essentially every vehicle and is the minimum requirement.
- **Elevation** follows §2.4. Without it, hill energy is omitted rather than guessed.
- **Vehicle mass and road-load coefficients** come from EPA data by make, model, and year, used only for the hidden sensor check (§3.5).

How the UI Labels It

Each trip carries a data-quality label: Measured, Estimated, Or Estimated from fill-ups. Breakdown lines that the available data cannot support are hidden, not filled with guesses. Tier C trips show a cost but no breakdown, since energy in was not measured for that trip.

3.5 Measured Trip Breakdown, Your History & Sensor Check

PowerTrip only shows numbers that were measured, or that follow from measured numbers and entered prices by plain arithmetic. It never shows modeled attributions such as “air drag cost \$0.23”, even labeled as estimates: they depend on wind, temperature, tires and load that the car does not report.

Where the Energy Went (Measured)

When the scanner logs battery power (pack voltage $\times$ current) and fuel rate for the whole drive, the trip's measured energy is summed over time and split several ways. Each split adds up to the trip total, and its cost uses the trip's own price per kWh and per gallon from the lot ledger (§3.1):

- **By speed:** city (under 40 mph, including stops), 40–60 mph, and over 60 mph. Miles, minutes, energy, cost and efficiency for each.
- **By terrain:** uphill, flat and downhill stretches (grade beyond $\pm 2\%$ over at least 60 m), when elevation is known (§2.4).
- **Stopped:** energy used below 0.5 mph (heat, A/C, electronics, idling).
- **Regen:** energy returned to the battery while no fuel is burning, in Wh per mile. Every car with a traction battery shows it, regular hybrids included, since their battery power is measured. Cars with a plug also show what it was worth at the trip's electricity price.
- **Fuel-free miles:** miles moving with no fuel burning (above 0.5 mph, fuel flow at most 0.05 gal/h). True on every car: all miles on an EV, electric miles on a plug-in hybrid, engine-off miles on a regular hybrid, and fuel-cut coasting on a gas car.

- **Electric vs. engine (plug-in hybrids):** miles and energy with the engine off and on, from the fuel-rate signal.

Downhill stretches can show net energy recovered. Vehicles without a complete power log get trip totals and the speed profile only; the breakdown is hidden rather than estimated.

Compared With Your Own Drives

Instead of theoretical what-ifs, the trip is compared with this car's own history. These are records of what happened, not causes: weather and traffic still differ between drives.

- **By speed band:** efficiency under 40, 40–60 and over 60 mph against the car's usual. All three bands always appear; a missing figure says why (No driving in this range on this trip, Too little to measure: 0.9 mi (needs 1 mi), No history yet: 1.9 mi from 3 drives (needs 3 mi)). A usual figure needs 3 mi from 2 or more drives.
- **Efficiency by speed:** a curve of measured efficiency in 2 mph steps across the car's drives, with this trip's own points on top. It counts only **steady cruising on flat road** (speed within 2 mph for 20 s, grade within 1% over 200 m), so time spent speeding up from stops does not make low speeds look inefficient. Each point shows how much driving it rests on; a point needs 1 mi from 2 or more drives, and gaps stay gaps rather than being smoothed over.
- **Most efficient steady speed:** the best point with 3 mi from 2 or more drives, shown only when equally well-measured speeds exist both slower and faster, so the slowest or fastest speed driven so far is never reported as the best.

Plug-in hybrids are compared separately for engine-off (mi/kWh) and engine-on (mpg) driving; an engine figure never rests on electric-only drives.

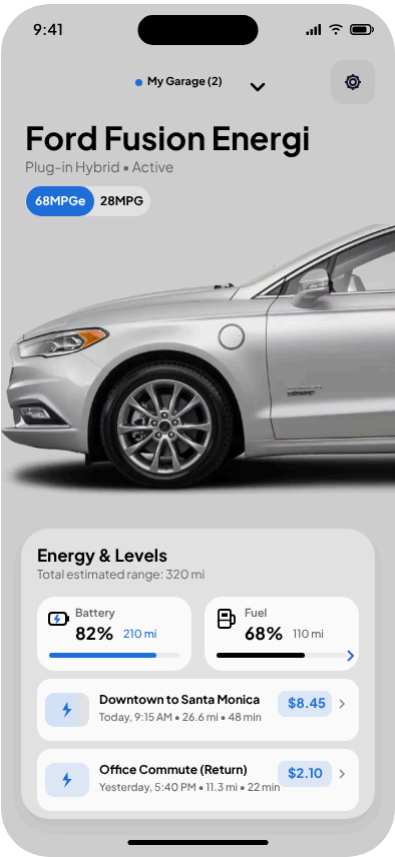
Hidden Sensor Check

Every trip's measured energy is compared with the work its speed, distance and hills required at the wheels, using the vehicle's EPA road-load coefficients (Test Car List "Target Coef" A, B, C and test weight). The ratio between the two is learned per vehicle from its own trips, separately for electric and engine driving. Nothing is shown while the car has fewer than 5 comparable trips, or while a trip is within 35% of the car's usual ratio. Beyond that, the trip shows one alert with the measured and expected amounts (This trip used 48% more electricity than your car usually needs for the same speed, distance and hills), since a large gap usually means a sensor problem or something physical: low tire pressure, a dragging brake, a roof box, strong wind or cold.

4. Interface Walkthrough & User Workflows

4.1 Home Screen: The Central Telematics Hub

The Home Screen serves as the primary gateway for all vehicle operations. It uses a button-driven architecture with zero persistent tab bars.

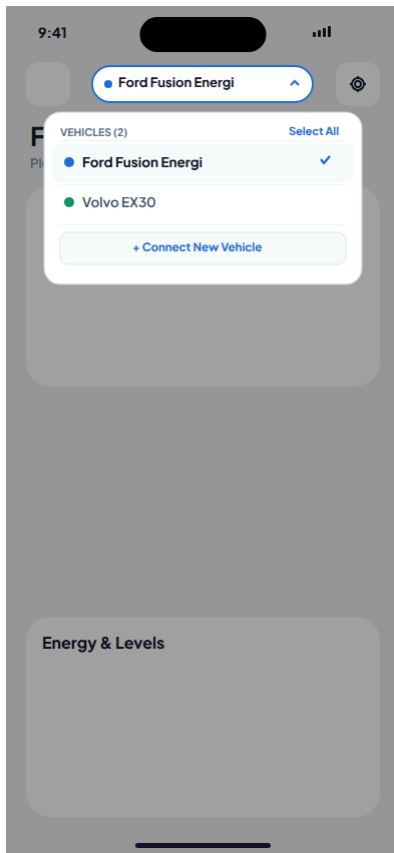


Interactive Touchpoint Architecture

- **Top Vehicle Pill (Ford Fusion Energi ▼):** Tapping the center pill opens the **Vehicle Selector Popover** to switch vehicles or activate Fleet mode.
- **Top-Right Squirrelle (⚙ Gear Icon):** Direct navigation touchpoint to the dedicated **Settings** page.
- **Energy & Levels Card:** Displays battery SoC% (82% • 17 mi) and gasoline tank level (68% • 380 mi). Tapping anywhere on this card navigates directly to **Spending & Expenses**.
- **Recent Drives List:** Displays the latest post-drive trips featuring the **Majority EV Squirrelle** avatar system. Tapping any drive card (>) navigates directly into that trip's **Trip Summary**.
- **Vehicle Photo / Subtitle:** Tapping the vehicle hero area navigates to **My Garage**.

4.2 Multi-Vehicle Selector Popover

Instead of an enterprise checkbox form or multi-page menu, the vehicle selector is an ultra-clean **Apple HIG-style floating popover**.

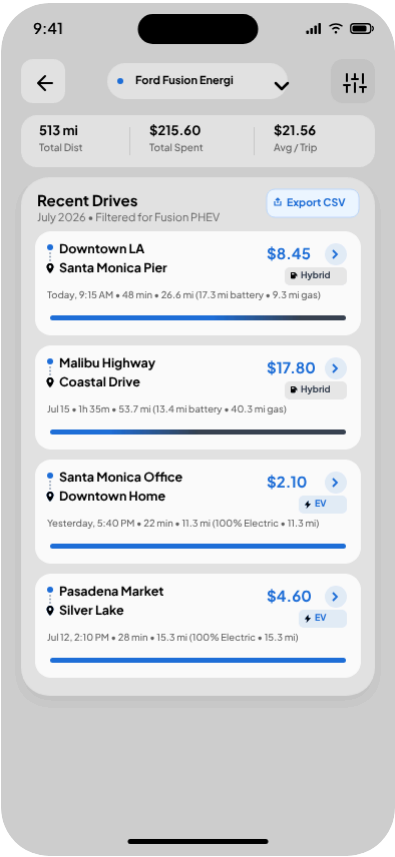


Native iOS Trailing Checkmark Selection

- **Border-Free Checkmarks:** Eliminates ugly square checkbox boxes in favor of clean vector checkmarks (✓) that only appear on active cars.
- **Minimal Data Density:** Displays strictly the vehicle's signature color dot and name (zero distracting fuel levels or specs).
- **1-Tap Multi-Car Filtering:**
 - **Single-Vehicle Mode:** Tap one car (e.g. **Ford Fusion Energi**). The header pill updates to that vehicle with its accent.
 - **Combined Fleet Mode:** Tap **Select All** (or select multiple vehicles). The header pill automatically updates to **●● All Vehicles (2) ▼**.
- **Direct Addition:** Includes a + Connect New Vehicle shortcut to pair another WiCAN scanner.

4.3 Trip History & GPS Timeline

Accessed via the recent drives touchpoint, Trip History logs all historical drives organized chronologically.

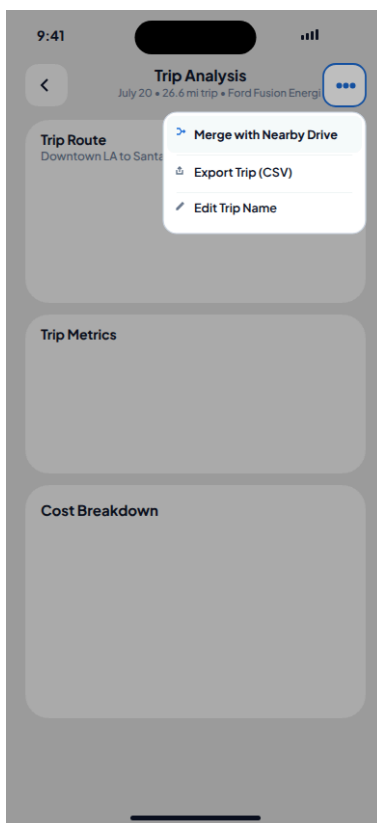
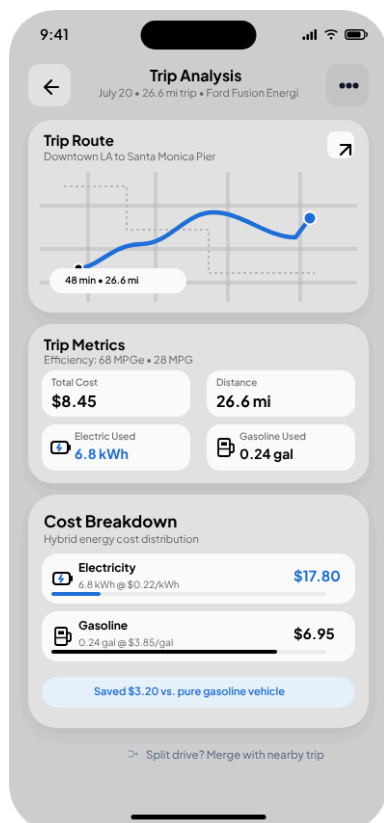


Solution 1: GPS Waypoint Timeline

- **Waypoint Inset Graphics:**
 - Departure: Solid accent dot ● in vehicle signature color.
 - Track: 3 discrete GPS navigation dashes (|).
 - Destination: Solid vector map pin (📍).
- **Dynamic Compact Heights:** Single-fuel cars omit bottom bars (compact 82px height). PHEV cards feature a full-width 295px dual-energy telemetry bar showing electric vs gas split.
- **Line 3 Financials:** Integrates dollar cost side-by-side with powertrain-specific efficiency metrics.
- **1-Tap Bulk Export:** Features a dedicated [↗ Export CSV] action pill in the header for instant data export.

4.4 Trip Analysis & De-Emphasized Merge Workflow

Tapping any trip card opens the deep post-drive analysis view.

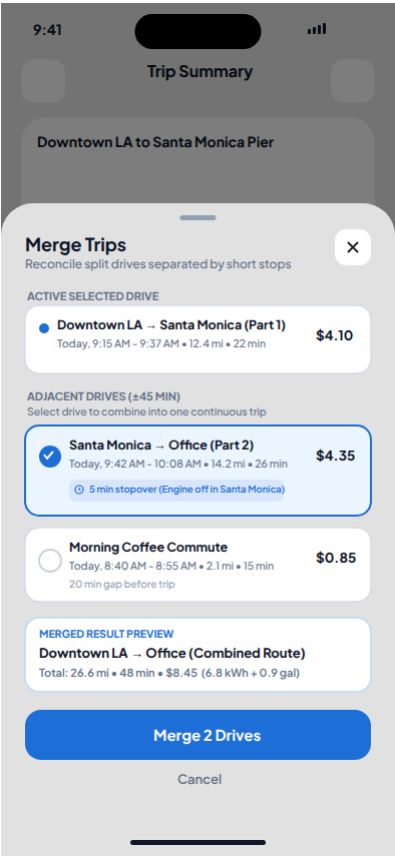


De-Emphasized Merge Action

- **Route Map & Telematics:** Visual elevation/speed profile, distance, duration, and dual energy consumption (kWh + gal).
- **Cost Breakdown:** Granular dollar split for electric utility cost vs gasoline burned.
- **Somewhat Hidden Action:** Because trip merging happens < 5% of the time, the loud button was removed:
 - **Header ... Menu:** Tapping the top-right squircle opens an iOS action popover with **Merge with Nearby Drive**, **Export CSV**, and **Edit Trip Name**.
 - **Discreet Footer Link:** Subtle, low-contrast text prompt: Split drive? Merge with nearby trip.

4.5 The Merge Trips Workflow

When WiCAN splits a drive due to a brief engine stop (e.g. running into a convenience store or stopping at a railroad crossing), the owner can merge both segments in two taps.

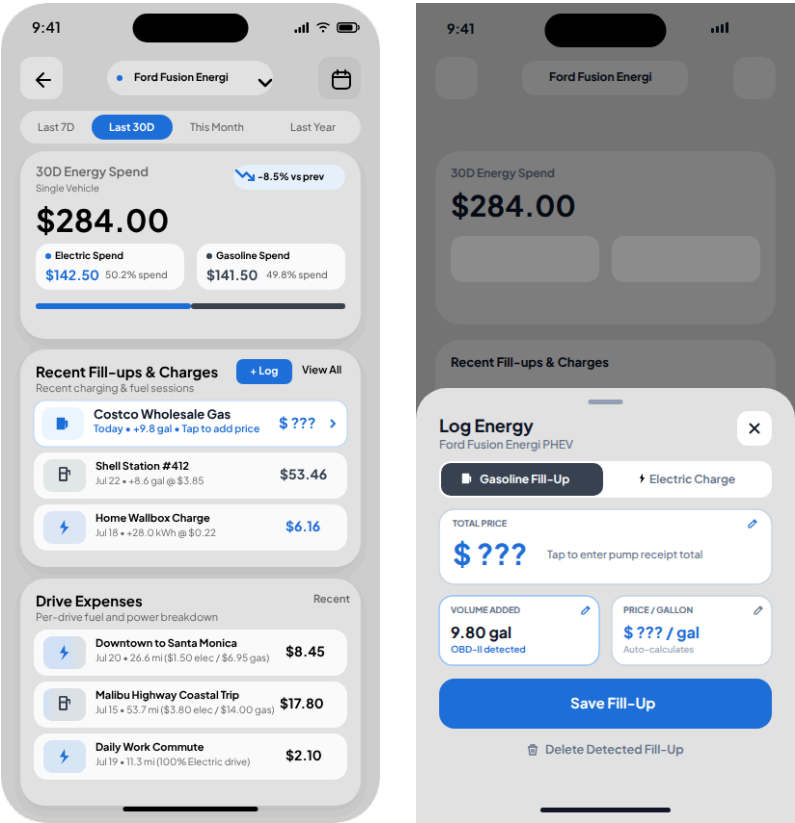


Intelligent Trip Reconciliation

- **Active Base Drive:** Displays the selected trip currently being analyzed (Downtown LA → Santa Monica • 12.4 mi • \$4.10).
- **Adjacent Drives Query (±45 min):** The server automatically queries drives that occurred within 45 minutes of the trip start or end.
- **Stopover Context Badge:** Highlights the exact stop duration: ⌚ 5 min stopover (Engine off in Santa Monica).
- **Combined Result Preview:** Real-time preview calculating merged route title, total mileage (26.6 mi), duration (48 min), cost (\$8.45), and combined telemetry (6.8 kWh + 0.9 gal).
- **Single Execution Button:** Primary Merge 2 Drives button executes atomic reconciliation on the server.

4.6 Spending & Refuel Tracking

Accessible directly by tapping the Energy & Levels card on the dashboard, this screen tracks energy economics.

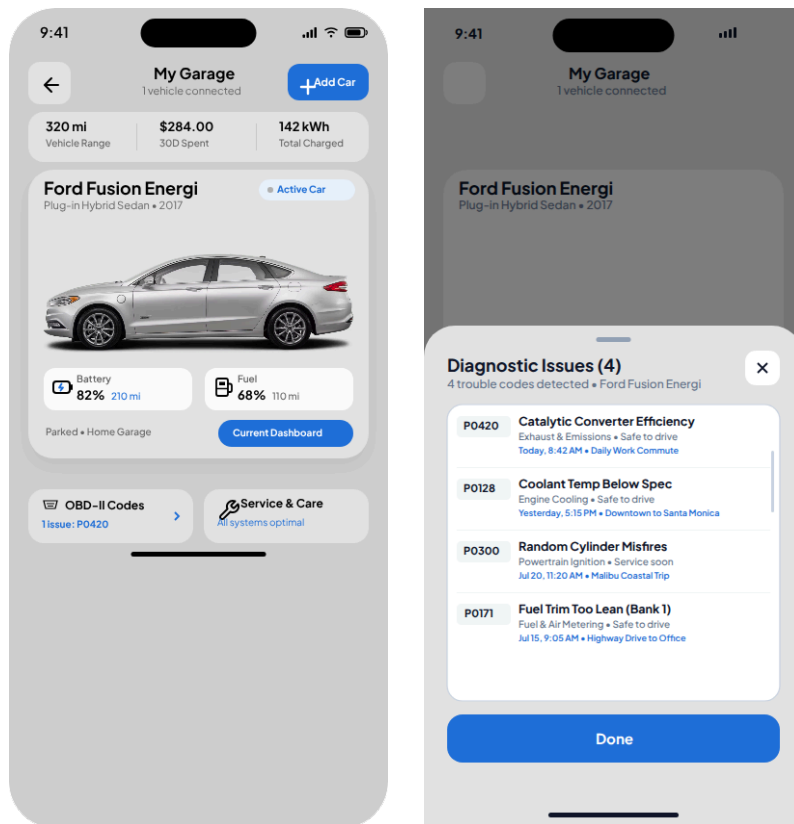


Minimal Refuel Logging

- **Fill-Up Detection:** When the fuel level rises while parked (\$3.1), an entry is created: Fill-up detected • Add details in Electric Blue, showing the gauge's approximate gallons.
- **Super Minimal Modal:** No odometer, no tank level, no location. Exactly two required fields from the receipt: Total Cost and Amount Added (gallons or kWh); Price/Gallon is calculated. For a detected fill-up, the amount is prefilled from the gauge and labeled as such.
- **Charging:** Home and free-place charges are measured by the scanner and priced automatically; only paid public charges need the modal.
- **Mode Switcher:** Toggle between Gasoline Fill-Up and Electric Charge.
- **Neutral Delete:** Calm slate action to delete false detections.

4.7 My Garage & Non-Alarming DTC Diagnostics

Accessible via the vehicle hero touchpoint, My Garage monitors overall vehicle health and diagnostic fault codes.

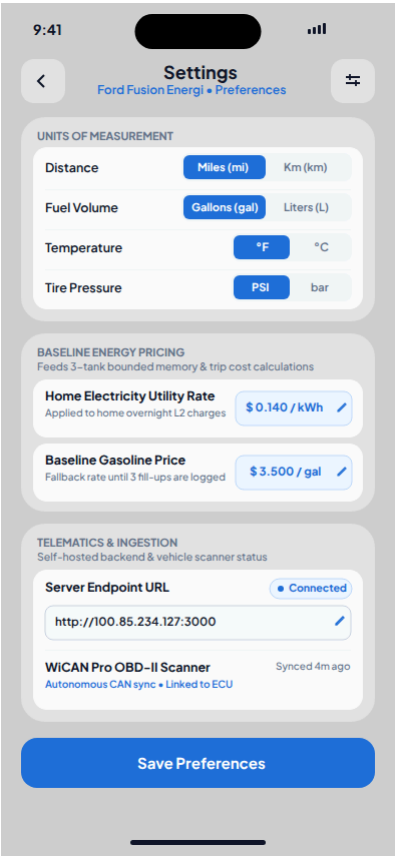


Owner-Centric Trouble Codes

- **Active Trigger State:** When a DTC is read, the OBD-II tile highlights with 1 issue: P0420 > in Electric Blue.
- **Plain-English Explanations:** Translates raw codes into owner-comprehensible terms: P0420 Catalytic Converter Efficiency • Exhaust & Emissions.
- **Safe-to-Drive Determination:** Informs the owner immediately whether the car is safe to drive or requires service soon.
- **Detection Timestamps & Trip Names:** Shows exact detection context: Read Today, 8:42 AM • During "Daily Work Commute".
- **Zero Clearing Clutter:** No clear code button (clearing without repair causes catalytic / inspection failure).

4.8 Dedicated App Settings Page

Accessed via the top-right gear icon (⚙️) on the Home Screen, the dedicated Settings page manages global app configurations.



Global Preferences & Server Ingestion

- **Units of Measurement:** Segmented pill controls for Distance (mi / km), Fuel Volume (gal / L), Temperature (°F / °C), and Tire Pressure (PSI / bar).
- **Baseline Energy Pricing:**
 - **Home Electricity Utility Rate** (\$0.220 / kWh): Feeds overnight home L2 charging calculations.
 - **Baseline Gasoline Price** (\$3.500 / gal): Last-resort price for unpriced fill-ups when no EIA regional price is available (\$3.1).
 - **Free Charging Locations:** Saved places (e.g. school, work) whose charging is priced at exactly \$0.00.
- **Editable Server Endpoint URL:** Input field (http://100.85.234.127:3000) with edit pencil to change the Docker backend host anytime.
- **Hardware Telematics Sync:** Real-time connection status for the WiCAN Pro scanner with last sync timestamp.
- **One Accent Rule Adherence:** Strict Electric Blue palette with zero generic green status pills.

5. Design System & Architectural Invariants

To guarantee visual harmony and prevent regression during future feature additions, all screens must strictly comply with the following invariants:

Design Parameter	Platform Standard & Enforcement
Canvas Foundation	#D1D1D1 canvas background. Container cards in #E3E3E3 (24–28px radius). Interactive sub-tiles in #FFFFFF 0.95 (12–14px radius). Never introduce full-page white scrims.
Single Accent Rule	Every single-car view must feature the neutral foundation paired with strictly ONE vehicle signature accent color : • Ford Fusion Energi: Electric Blue (#1E6FD9) • Volvo EX30: Emerald Green (#0C9660) • Toyota RAV4 Hybrid: Warm Amber (#D97706) • Volvo XC90 B6: Slate Charcoal (#374151)
No Generic Green/Red	Never use generic “Green = Good / Red = Bad” status colors that clash with vehicle branding. Normal tire pressure, 0 DTC faults, and online server status must use calm neutral slate or the vehicle’s signature accent color.
Vector Geometry Only	Zero emojis or text glyphs (✂️, 🗑️, 📧, ✖️, ▼, 🚫). All icons must be rendered using vector SVG paths (<code>penpot.createPath()</code>) or stroked geometry.
No Blinking Cursors	Never render mock blinking text cursor artifacts () in inputs. Use edit pencil icons (✎), subtle container strokes, and placeholder text.
Integer Radii	Penpot API throws errors on non-integer border radii. Always enforce integer values (<code>Math.round()</code>).
Hierarchical Navigation	Zero bottom tab bars. Navigation flows organically via direct card touchpoints, with standard 44×44px squircle back buttons (<) on all sub-screens.

6. Implementation Roadmap

The Penpot designs define the core screens; they continue to be refined alongside development.

- Phase 1: WiCAN Pro Firmware & Ingestion:**
 - Flash WiCAN Pro with autonomous CAN logging firmware.
 - Configure circular SPI flash buffer and deep sleep wake triggers (> 13.1V bus activity).
 - Configure home Wi-Fi SSID and MQTT/HTTP upload payload specs.
 - Sniff the reference Fusion Energi to confirm broadcast GPS frames and the real request identifiers for motor torque, brake pressure, and cabin heater power.
- Phase 2: Vehicle Profiles & Docker Backend:**
 - Deploy Home Telematics Hub Docker container with PostgreSQL database.
 - Build the vehicle profile format and the generic OBD-II profile (§2.3, layer 1); add the verified Fusion Energi profile.
 - Implement the Energy Lot Ledger (§3.1) with the reconciliation invariant as an automated test.
 - Implement data availability tiers A–C (§3.4).
- Phase 3: Frontend Client Integration:**
 - Continue the Android client in the PowerTrip-Android repository (Kotlin + Jetpack Compose), implementing the Penpot design tokens and SVG assets documented in this specification.
 - Build the web interface, served by the home server, from the Penpot “Web Desktop” designs.

4. Phase 4: Expansion:

- EPA road-load lookup by VIN (NHTSA vPIC decode, EPA Test Car List match) for the sensor check (§3.5).
- Additional vehicles from community poll lists and opendbc (§2.3), each verified before release.
- Phone relay for non-WiCAN scanners (§2.5).
- Regions beyond the US (e.g. EU): regional fuel price feeds, and road-load defaults where EPA coefficients are unavailable (e.g. from WLTP data or per-vehicle-class averages).

— End of Automotive Telematics Platform Specification Document —